

DPA-Resistant QDI Dual-Rail AES S-Box Based on Power-Balanced Weak-Conditioned Half-Buffer

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Abstract— We propose an asynchronous-logic (async) Quasi-Delay-Insensitive (QDI) dual-rail 32-bit Advanced Encryption Standard (AES) Substitution-Box (S-Box) for Differential Power Analysis (DPA) attack countermeasure. There are three novel features in the proposed S-Box. First, the proposed S-Box operates in async QDI protocol with dual-rail data encoding, hence there is only a marginal difference in power dissipation for different signal output transitions. Second, the proposed S-Box embodies the power-balanced async Weak-Conditioned Half-Buffer (WCHB) cell approach, which features the same number of transitions, and hence same number of switching for different input combinations to equalize the power dissipation. Third, the proposed S-Box embodies our novel-designed library cells in which each output wire, corresponding to different transitions, has a similar capacitive load, hence hiding the dynamic power dissipation. Based on the 65nm CMOS process, we implement the proposed 32-bit AES S-Box, and benchmark it against the conventional synchronous-logic (sync) S-Box and the reported async (i.e. unbalanced) WCHB S-Box. From the simulation results, our proposed power-balanced WCHB S-Box features significantly lower 1.21% and 0.55% of Normalized Energy Deviation (NED) and Normalized Standard Deviation (NSD) respectively. Particularly, our proposed design is of 57.6× and 17.9× lower NED, and 55.1× and 9.29× lower NSD than the reported sync and async counterparts respectively.

Keywords—Power-Balanced, Weak-Conditioned Half-Buffer, Differential Power Analysis, Advanced Encryption Standard

I. INTRODUCTION

Cryptography is aimed to protect and ensure the security of cipher by performing a series of encryption operations with the secret key to conceal the plaintext. The use of cryptography is widely embraced to secure electronic systems such as personal digital assistants, mobile phones, and smart cards [1] in many applications including defence and banking. Advanced Encryption Standard (AES) is the most commonly adopted encryption algorithm due to its higher reliability against cybercrimes and data leakages [2]. However, the AES Substitution-Box (S-Box), is vulnerable against Side-Channel Attacks (SCAs) [3]. This is due to the fact that attackers can exploit the correlation between the processed data during the encryption process and the corresponding Physical Leakage Information (PLI) such as power dissipation, operation timing, and electromagnetic (EM) emission [1] to reveal the secret key.

Differential Power Analysis (DPA) attack is one of the most straightforward and effective SCAs that is conducted in non-invasive and affordable off-the-shelf equipment [3]. To countermeasure against the DPA attack, the power profile of cryptographic devices should be equalized (or flattened) in order to prevent the attack.

However, in the commercial synchronous-logic (sync) design, the single-rail data are propagated and latched by the clock. The power dissipation could be easily differentiated for

various input combinations and various signal output transitions. Further, the data operation time could be easily identified as the operation often starts near the clock edge. In view of these, asynchronous-logic (async) [4] can be applied to modulate the power profile amplitude and operation time as the async data are encoded in dual-rail modality and propagated in handshaking protocol, e.g. Quasi-Delay-Insensitive (QDI) timing model [4].

Several SCA countermeasures have been reported to realize the AES S-Box by using dual-rail modality including Sense Amplifier Based Logic (SABL) [5], Wave Dynamic Differential Logic (WDDL) [6], Masked Dual-rail Pre-charge Logic (MDPL) [7], Dual-rail Random Switching Logic (DRSL) [8], and Weak-Conditioned Half-Buffer (WCHB) [9]. Of these reported circuit templates, SABL was reported to have a low correlated PLI with the associate signal transitions. However, its circuit is rather complex (i.e. non-standard cell based) and requires a new core cell library. As SABL's refinement, K. Tiri *et al.* [6] proposed WDDL which utilizes rudimentary cells such as AND and OR gates but at the expense of larger area, power dissipation and longer delay. Based on Stefan Mangard *et al.* [11], WDDL can be attacked by injecting glitches to a masked AES hardware circuit. In order to circumvent these shortcomings, MDPL [7] was proposed to mask the intermediate value with randomized bits. Although it is better, it somewhat remains error-prone against the glitch attack. Subsequently, Chen *et al.* [8] proposed DRSL in which all inputs of every gates are synchronized with the pre-charge signal to minimize the power dissipation in the presence of glitches at the expense of larger circuit area. Conversely, WCHB [9] employs a weak-conditioned inductability to countermeasure against the glitches attack. This was done by employing only the OR gates and C-Muller gates in constructing all the minterms Boolean logic functions for the corresponding cell outputs. However, the imbalance number of gates and the use of standard OR gates may still leak out the PLI through switching power.

In this paper, we propose an async QDI dual-rail S-Box to countermeasure against the DPA attack. There are three novel features in this approach. First, we adopt async QDI dual-rail handshake protocol to balance the power dissipation for different output transitions. Second, we employ the power-balanced WCHB approach, which realizes the same number of transitions for different input combinations to equalize the power dissipation. Third, we modify the standard gates so that each transition has the similar capacitive load, making the dynamic power dissipation to be independent from the switching wires. Based on the 65nm CMOS process, we realize the proposed S-Box and compare it against the reported async WCHB and sync counterparts using Normalized Energy Deviation (NED) and Normalized Standard Deviation (NSD) parameters. Based on the simulation results, our proposed power-balanced WCHB-based S-Box has the lowest NED of

1.21% and the lowest NSD of 0.55%. In short, our proposed design features $57.6\times$ and $17.9\times$ lower NED than the reported sync and async counterparts respectively. Furthermore, our proposed design is of $55.1\times$ and $9.29\times$ lower NSD than the reported sync and async counterparts respectively

This paper is organized as follows. Section II elaborates our proposed S-Box, power-balanced WCHB and library cells characterization. Section III discusses the simulation results. Finally, conclusions are drawn in Section IV.

II. PROPOSED ASYNC QDI DUAL-RAIL S-BOX

A. S-Box Architecture

We first design the AES S-Box with wordlength of 32-bit, async QDI handshake protocol and dual-rail data encoding. In overview, Fig. 1(a) depicts the AES-128 encryption algorithm, which consists of the AddRoundKey, SubBytes (S-Box), ShiftRows and MixColumns blocks, to encrypt plaintexts to ciphertexts. In the AES-128 encryption implementation, we focus on the S-Box architecture as it dominates the power dissipation, i.e. more than 80% in the AES encryption and hence leaking out the PLI.

Fig. 1(b) depicts the S-Box architecture that is adopted from the literature [10], processing 32-bit input data D_{in} to output data D_{out} . In the 32-bit S-Box, the straight line and the dot line represents the encryption process and decryption process of the async QDI dual-rail data respectively. Single-to-

dual rail converter and dual-to-single rail converter are used to switch the data signals between single-rail encoding (external interface) and dual-rail encoding (internal interface).

For the encryption process, D_{in} will mainly pass through multiplicative version in $GF(2^8)$ and affine transformation [10] to be evaluated to D_{out} . Fig. 1(c) depicts the multiplicative version in $GF(2^8)$ that follows the procedures as follows. First, map operation converts the 8-bit input into elements of $GF(2^4)$, i.e. a^h and a^l . Second, the $GF(2^4)$ elements are multiplied in square and followed by a modular reduction. Third, a series of multiplication and XOR operations extend the field of $GF(2^4)$ to the field of $GF(2^8)$. Fig. 1(d) depicts the affine transformation that follows a series of XOR and inverter operations, where i and q represents the 8-bit input and output respectively.

From the architecture view, the async QDI dual-rail S-Box switches the dual-rail output between empty state and valid state for alternate evaluate and precharge operations. In short, the design would switch the dual-rail output for all signal output transitions, i.e. 0-to-0, 0-to-1, 1-to-0 and 1-to-1, making the lesser difference in power dissipation. This is different from the conventional sync design, which only switches the output state if the output values are different from the previous values, i.e. 0-to-1 and 1-to-0 signal output transitions. There are no switchings for the 0-to-0 and 1-to-1 signal output transitions.

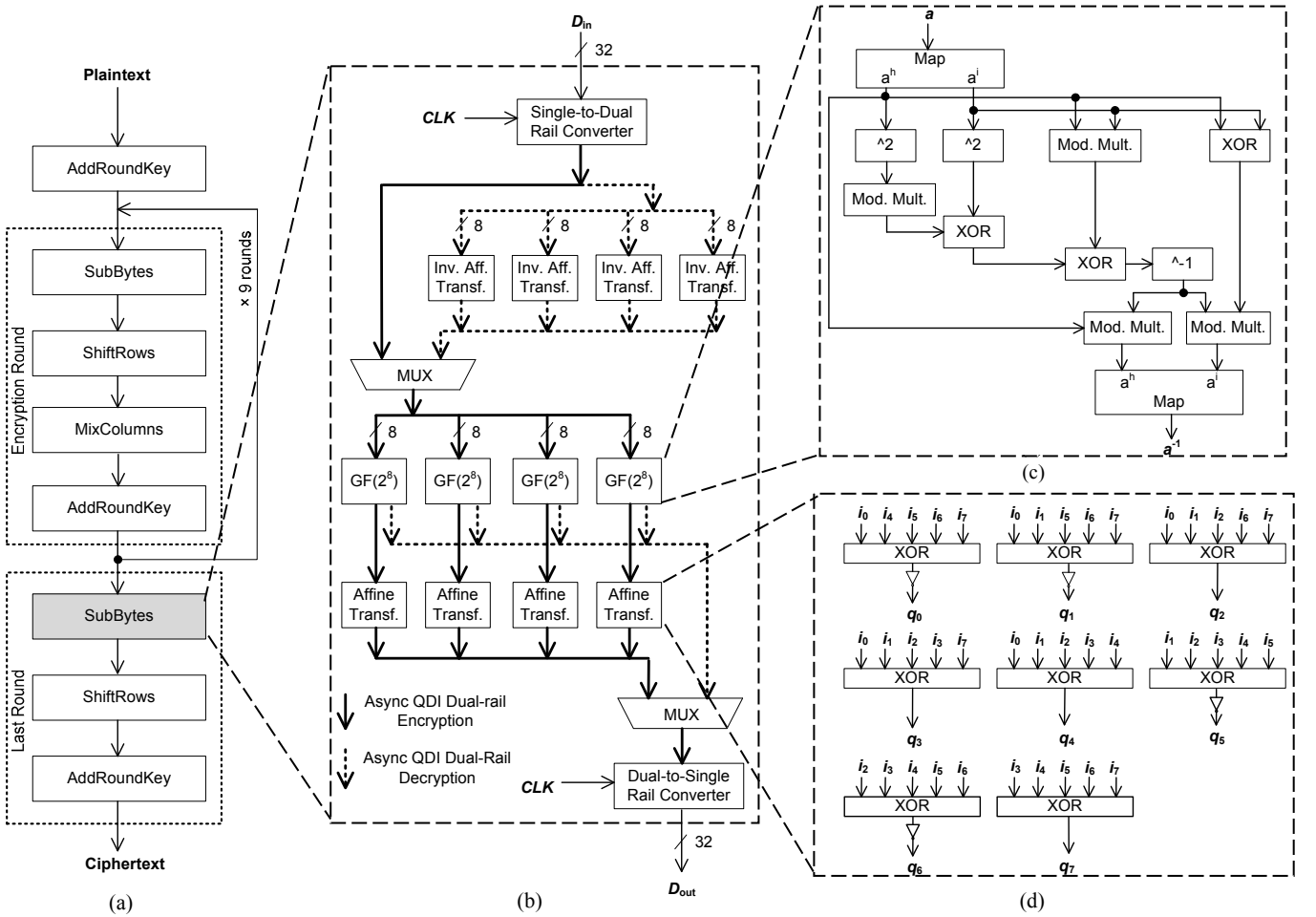


Fig. 1: (a) AES Encryption Implementation, (b) 32-bit Async QDI Dual-Rail S-Box, (c) 8-bit Multiplicative Inversion in $GF(2^8)$ and (d) 8-bit Affine Transformation

B. Power-Balanced WCHB Approach

In the async QDI dual-rail 32-bit S-Box, we propose a power-balanced WCHB cell approach that significantly reduces the difference in power dissipation for different input combination. For sake of brevity, we use 2-input AND/NAND cell for explanation. Fig. 2 depicts the proposed power-balanced WCHB 2-input AND/NAND cell, which propagates input signals, $A.T/A.F$ and $B.T/B.F$ to output signals, $Q.T/Q.F$, accompanied by the handshake signals $Rack/Lack$. The logical function is expressed in equations (1) and (2). In Fig. 2, the proposed WCHB 2-input AND/NAND cell consists of an input completion detection (ICD), a functional block, a latch and an output completion detection (OCD). The function of each block could be referred to [9].

$$Q.T = A.T \cdot B.T \quad (1)$$

$$Q.F = A.F \cdot B.T + A.T \cdot B.F + A.F \cdot B.F \quad (2)$$

In terms of the difference in power dissipation for various input combination, we elaborate the advantages of our proposed power-balanced WCHB over the reported WCHB [9]. There are 4 input combination for $AT \cdot BT$, i.e. 00, 01, 10 and 11. In the ICD, two more C-Muller gates are added to connect to an additional 3-input OR gate in the functional block for balancing the number of gates. Only one of the six C-Muller gates will be switched for every input combination. In the functional block, one of the two 3-input OR gates will be switched for all input combination. Hence there is a lesser difference in switching power for all input combination, i.e. 00/01/10/11.

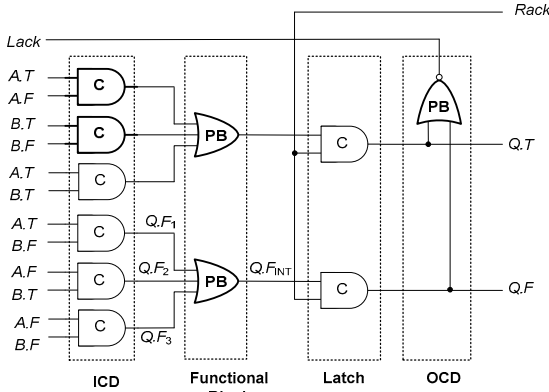


Fig. 2: Proposed Power-Balanced WCHB 2-Input AND/NAND Cell

Further, the 3-input OR gates are improved to be power balanced, as depicted in Fig. 3(a). For input combination 00/01/10, the intermediate signals $Q.F_1$, $Q.F_2$ and $Q.F_3$ will switch similar power. When $Q.F_1 = 1$, $(C_2 + 2C_3)$ are charged and $(2C_1 + C_2 + C_F)$ are discharged. When $Q.F_2 = 1$, $(2C_1 + C_3)$ are charged and $(2C_2 + C_3 + C_F)$ are discharged. When $Q.F_3 = 1$, $(C_1 + 2C_2)$ are charged and $(C_1 + 2C_3 + C_F)$ are

discharged. In short, there is the same number of charge/discharge capacitances for $Q.F_1 = 1$, $Q.F_2 = 1$ and $Q.F_3 = 1$.

In the latch, only one of the two C-Muller gates will be switched for every input combination. In the OCD, the output signals $Q.T$ and $Q.F$ will switch similar power due to the power-balanced 2-input NOR gate, as illustrated in Fig. 3(b). When $Q.T = 1$, C_F is charged and $(C_T + C_L)$ are discharged. Whereas when $Q.F = 1$, C_T is charged and $(C_F + C_L)$ is discharged. In short, there is the same number of charge/discharge capacitances for $Q.T = 1$ and $Q.F = 1$. All the above power balancing effects collectively make the power dissipation more difficult to be differentiated for various input combination.

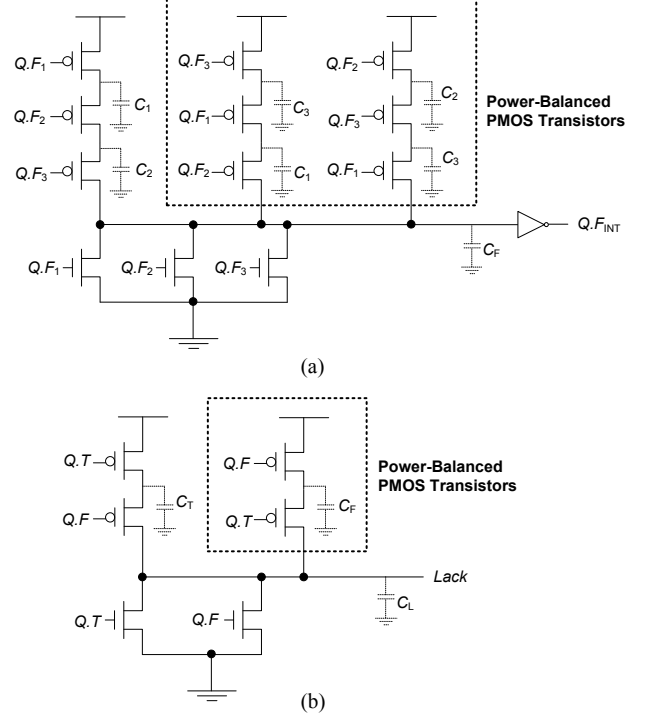


Fig. 3: Proposed Power-Balanced Standard Gates (a) 3-Input OR and (b) 2-Input NOR

C. Characterization of Library Cells

We further characterize the library cells embodying our proposed power-balanced WCHB, as tabulated in Table I based on 65nm CMOS process at 1.2 V and 27°C. In the characterization, we use the simulation parameters as follows: transistor count, cycle delay, maximum energy dissipation, minimum energy dissipation, NED and NSD. Transistor count is the number of transistors used to build the library cells. Cycle delay is the shortest time duration to complete evaluate and precharge operation for one cycle. Energy dissipation is the power dissipated for one operation cycle. Maximum energy dissipation and minimum energy dissipation are obtained by using different input combination. NED is

TABLE I. COMPARISON OF LIBRARY CELLS BETWEEN PROPOSED POWER-BALANCED WCHB AND REPORTED WCHB @ 65NM CMOS, 1.2V, 27°C

Library Cells	Transistor Count		Cycle Delay (ns)		Max Energy (fJ)		Min Energy (fJ)		NED (%)		NSD (%)	
	Reported	Proposed	Reported	Proposed	Reported	Proposed	Reported	Proposed	Reported	Proposed	Reported	Proposed
2-input AND/NAND	88	134	0.62	0.92	46.7	46.4	38.3	43.6	18.0	6.25	9.98	3.13
2-input OR/NOR	88	134	0.69	1.05	53.1	49.1	37.0	46.1	30.4	5.89	15.1	2.99
3-input AND/NAND	184	394	1.52	1.41	99.5	131	85.8	121.2	13.7	7.19	4.67	3.01
3-input OR/NOR	184	394	1.56	1.74	102	131	84.3	124	17.9	5.15	6.15	2.46
3-input AO/AOI	144	234	0.91	1.50	76.4	79.3	60.2	77.1	21.2	2.71	8.54	1.40
3-input OA/OAI	144	234	0.91	1.49	74.9	78.5	60.2	76.7	19.6	2.27	7.37	1.19
Average	139 (0.55×)	254 (1×)	1.04 (0.77×)	1.35 (1×)	75.4 (0.88×)	85.9 (1×)	61 (0.75×)	81.5 (1×)	20.1 (4.1×)	4.91 (1×)	8.64 (5.7×)	1.51 (1×)

expressed in equation (3). NSD is defined as standard deviation of energy dissipation per cycle divided by its mean.

$$NED = \frac{Max(energy\ per\ cycle) - Min(energy\ per\ cycle)}{Max(energy\ per\ cycle)} \quad (3)$$

From Table I, despite the higher transistor count, longer cycle delay and higher energy dissipation, the proposed power-balanced WCHB library cells features, on average, $4.1\times$ lower NED and $5.7\times$ lower NSD than the reported WCHB library cells. The higher overheads are due to the additional C-Muller and OR gates for power balancing, and the additional power-balanced PMOS transistors in the standard gates. Nonetheless, these design efforts make the power dissipation to be balanced for various input combination, i.e. lower NED and lower NSD, where the power dissipation is more difficult to be differentiated in the DPA attack.

III. SIMULATION RESULTS

Based on 65nm CMOS process, we realize our proposed power-balanced WCHB cells e.g. 2-input AND/NAND, 2-input OR/NOR, 3-input AND/NAND, 3-input OR/NOR, 3-input AO/AOI and 3-input OA/OAI, in the 32-bit AES S-Box. In the S-Box realization, we use Verilog logic model by designing the basic library cells using Cadence Virtuoso, and run the simulation using Synopsys Nanosim. We benchmark the power-balanced WCHB AES S-Box against the conventional sync S-Box and the reported (i.e. unbalanced) WCHB S-Box. We further compare our design with the reported S-Boxes embodying various cell approaches and employing various power balancing techniques, as tabulated in Table II. In general, the dual-rail data encoding is widely adopted to balance the power dissipation for different output transitions. The async designs generally suffers from higher area/gate count/transistor count overhead than the sync designs due to the additional acknowledgement circuitry for data propagation. It is observed that our design does not have a significant improvement in terms of frequency and energy dissipation. In the security applications, the main focus is to maintain the power balancing and hence hiding the PLI and thus the speed and the energy dissipation are not of the key concern. Therefore, NED and NSD are the most important parameters used to quantify the resistant level of the circuit security as lower values of NED and NSD would require the attackers to use more complex techniques and more expensive equipment to statistically analyze the power consumption of cryptographic devices.

Note that the proposed S-Box features only a marginal difference in power dissipation for different output transitions and input combination. This is due to the attractive features of our proposed async QDI dual-rail in the S-Box architecture and the power-balancing feature in the WCHB cell approach. Overall, our proposed power-balanced WCHB S-Box features

amongst the lowest NED of 1.2% and NSD of 0.55% as compared to other designs. For benchmarking, as compared to our proposed power-balanced WCHB S-Box, the conventional sync S-Box has $57.6\times$ higher NED and $55.1\times$ higher NSD, while the reported (i.e. unbalanced) WCHB S-Box has $17.9\times$ higher NED and $9.29\times$ higher NSD.

IV. CONCLUSIONS

In this paper, we have proposed a DPA-resistant 32-bit AES S-Box with async QDI dual-rail handshaking protocol and based on the power-balanced WCHB cell approach. Using 65nm CMOS process technology, we have implemented the proposed S-Box, and benchmarked it against the reported WCHB and sync counterparts. Our proposed power-balanced WCHB-based S-Box features the lowest 1.21% of NED and 0.55% of NSD amongst three designs, and is comparable to the reported state-of-the-art S-Box designs. The reported sync counterpart is of $57.6\times$ higher NED and $55.1\times$ higher NSD, while the unbalanced WCHB-based S-Box is of $17.9\times$ higher NED and $9.29\times$ higher NSD than our proposed power-balanced WCHB-based S-Box design.

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TABLE II. SIMULATION RESULTS OF VARIOUS S-BOXES NORMALIZED TO THE READINGS OF PROPOSED WCHB S-BOX

S-Box	CMOS	Operation Modality	Data Encoding	Power-balancing Techniques	Frequency	Energy	NED	NSD
SABL [5]	180nm	Sync	Dual-rail	Sense amplifier and balance logic	-	$0.69\times$	$2.64\times$	$1.09\times$
Conventional Sync	65nm	Sync	Single-rail	None	$0.95\times$	$0.01\times$	$57.6\times$	$55.1\times$
Reported WCHB	65nm	Async	Dual-rail	None	$1.27\times$	$1.10\times$	$17.9\times$	$9.29\times$
Proposed WCHB	65nm	Async	Dual-rail	Balanced gates # + power-balanced std gates	$1\times$ (105MHz)	$1\times$ (163,000fJ)	$1\times$ (1.21%)	$1\times$ (0.55%)